

Fixed Point Structure of Gradient Flow Exact Renormalization Group

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2021/12/8 @ KEK-TH 2021

Based on work in progress with Y.Hamada(KEK) and Y.Abe

Exact Renormalization Group

- A framework to study physics under **varying the energy scale**
- The change of the effective action S_τ is described by **a differential equation** → “ERG equation”
- A typical example : Wilson-Polchinski equation

$$\partial_\tau S_\tau = \int_p \left\{ \left[\left(2p^2 + \frac{D+2}{2} - \frac{\eta_\tau}{2} \right) \phi_i(p) + p_\mu \frac{\partial}{\partial p_\mu} \phi_i(p) \right] \frac{\delta S_\tau}{\delta \phi_i(p)} + \frac{1}{p^2} \left[4p^2 k(p) + 2p^2 \frac{dk(p)}{dp^2} - \eta_\tau k(p) \right] \frac{1}{2} \left(\frac{\delta^2 S_\tau}{\delta \phi_i(p) \delta \phi_i(-p)} - \frac{\delta S_\tau}{\delta \phi_i(p)} \frac{\delta S_\tau}{\delta \phi_i(-p)} \right) \right\}$$

$$\partial_\tau \text{ (circle with } S_\tau \text{)} = \text{ (circle with } S_\tau \text{)} + \text{ (circle with } S_\tau \text{ and a self-loop)} - \text{ (circle with } S_\tau \text{) --- (circle with } S_\tau \text{)}$$

$(\Lambda = \mu e^{-\tau})$

Gradient Flow

- A method to construct **composite operators without the equal-point singularity**

[M.Lüscher, P.Weisz 1101.0963]

- Gradient flow equation for the linear sigma model
(the diffusion equation)

$$\partial_t \varphi(t, x) = \partial_x^2 \varphi(t, x), \quad \varphi(0, x) = \phi(x)$$

[F. Capponi, L. Debbio, S. Ehret, R. Pellegrini, A. Rago 1512.02851]

- Correlation functions are **UV finite** with wave function renormalization:

$$Z_t^{-n/2} \langle \varphi(t_1, x_1) \varphi(t_2, x_2) \cdots \varphi(t_n, x_n) \rangle_\phi < \infty$$

even for the equal point case (e.g. $x_1 = x_2$)

Gradient Flow ERG (GFERG)

- The gradient flow resembles to the coarse graining along renormalization group flows
- If we get the solution $\varphi'(t, x)$ to a GF eq. with $\varphi'_i(0, x) = \phi'_i(x)$, Wilson action can be defined as

[H. Sonoda and H. Suzuki 2012.03568]

$$e^{-S_\tau[\phi_i]} := \exp \left[\int_{x,y} \frac{1}{2} \mathcal{D}(x-y) \frac{\delta^2}{\delta \phi_i(x) \delta \phi_i(y)} \right] \\ \times \int [D\phi'_i] \prod_{x',i} \delta \left(\phi_i(x) - e^{\tau(D-2)} Z_\tau^{1/2} \varphi'_i(t, x' e^\tau) \right) \\ \times \exp \left[- \int_{x'',y''} \frac{1}{2} \mathcal{D}(x''-y'') \frac{\delta^2}{\delta \phi'_i(x'') \delta \phi'_i(y'')} \right] e^{-S_{\tau=0}[\phi'_i]}$$

where $t := e^{2\tau} - 1$

$$\mathcal{D}(x) := \int_p e^{ipx} \frac{k(p)}{p^2}, \quad k(p) \cdots \text{cutoff function } (k(\infty) = 0)$$

Gradient Flow ERG (GFERG)

- If we choose φ'_i as the solution to the diffusion eq., S_τ satisfies the WP eq.
- Correlation functions of ϕ_i are UV finite
 $\Rightarrow S_\tau$ has UV regularization
- S_τ for gauge fields is invariant under gauge transformation

[H. Sonoda and H. Suzuki 2012.03568]

$$\begin{aligned}
 e^{-S_\tau[A_\mu^a]} := & \exp \left[\int_{x,y} \frac{1}{2} \mathcal{D}(x-y) \frac{\delta^2}{\delta A_\mu^a(x) \delta A_\mu^a(y)} \right] \\
 & \times \int [DA'_\mu] \prod_{x',i} \delta \left(A_\mu^a(x) - e^{\tau(D-4)} B'^a_\mu(t, x' e^\tau) \right) \\
 & \times \exp \left[- \int_{x'',y''} \frac{1}{2} \mathcal{D}(x''-y'') \frac{\delta^2}{\delta A'_\mu^a(x'') \delta A'_\mu^a(y'')} \right] e^{-S_{\tau=0}[A'^a_\mu]}
 \end{aligned}$$

where $\partial_t B'_\mu = D'_\nu G'_{\nu\mu} + \alpha_0 D'_\mu \partial_\nu B'_\nu$ with $B'_\mu(0, x) = A'_\mu$

Our motivation

- Gradient flow eq. for general scalar models can be different from the diffusion eq.
e.g) Non-linear sigma model

$$\partial_t \varphi_i = \partial_x^2 \varphi_i - (\varphi_j \partial_x^2 \varphi_j) \varphi_i$$

[H.Makino, H.Suzuki 1410.7538]

- GFERG equation depends on the GF eq.
It differs from the WP eq. in general cases
- Is GFERG consistent with conventional ERG?
e.g.) fixed points, critical exponents

Our work

- We study the fixed points of the general GFERG eq. for scalar fields
→ The fixed points of the WP eq. are realized in the $\tau \rightarrow \infty$ limit along the GFERG flow
- We show the GFERG eq. has a similar RG flow structure around a fixed point to the WP eq.
→ Scaling dimensions of relevant or marginal operators are the same, while those of irrelevant operators can be different
- Non-linear terms in the GF eq. do not affect the fixed point structure in the $\tau \rightarrow \infty$ limit

GFERG with general GF

- general GF equation

$$\partial_\tau \phi_i = \underbrace{\partial_x^2 \phi_i}_{\text{WP part}} + \underbrace{\sum_{n=2}^{\infty} \int_{x_1, \dots, x_n} f_i^{i_1, \dots, i_n}(x; x_1, \dots, x_n; \partial_{x_1}, \dots, \partial_{x_n}) \phi_{i_1}(\tau, x_1) \cdots \phi_{i_n}(\tau, x_n)}_{\text{extra terms}}$$

- GFERG equation

$$\begin{aligned} \partial_\tau e^{-S_\tau[\Phi]} &= (\text{WP part}) \\ &+ \sum_{n=2}^{\infty} \lambda^{n-1}(\tau) \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_i^{i_1, \dots, i_n} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right) \times \cdots \right. \\ &\quad \left. \times \left(\phi_{i_n}(x_n) + \int_{y_n} \mathcal{D}(x_n - y_n) \frac{\delta}{\delta \phi_{i_n}(y_n)} \right) \right\} e^{-S_\tau[\Phi]} \end{aligned}$$

where $\lambda(\tau) := \exp\left(-\tau \frac{(D-2)}{2}\right) Z_\tau^{1/2}$

Fixed Point action

- Fixed point action S^* is defined as

$$\partial_\tau S^* = 0 \quad \text{and} \quad S^* := \lim_{\tau \rightarrow \infty} S_\tau$$

- S^* satisfies

$$0 = (\text{WP part}) + \sum_{n=2}^{\infty} \lambda^{n-1}(\infty) \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_i^{i_1, \dots, i_n} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right) \times \dots \right. \\ \left. \times \left(\phi_{i_n}(x_n) + \int_{y_n} \mathcal{D}(x_n - y_n) \frac{\delta}{\delta \phi_{i_n}(y_n)} \right) \right\} e^{-S^*[\Phi]}$$

- Asymptotic behavior of $\lambda(\tau)$

$$\lambda(\tau) \sim \exp(-\tau(D - 2 + \eta)/2) \quad \frac{\eta}{2} := \frac{d}{d\tau} \log Z_\tau \Big|_{\tau=\infty}$$

- $\lambda(\infty)$ should vanish ($\rightarrow$ Next slide)

Vanishing λ

- Connected two point function at the fixed point theory

$$\langle \phi(x)\phi(0) \rangle_{\text{connected}} \propto \frac{1}{x^{D-2+\eta}}$$

- This should become zero when $|x| \rightarrow \infty$
 - $\rightarrow D - 2 + \eta > 0$ is required
 - $\rightarrow \lambda(\infty)$ ($\lambda(\tau) \sim e^{-\tau(D-2+\eta)/2}$) and the extra terms vanish and only the WP part survives
 - $\Rightarrow S^*$ satisfies the fixed point condition of the WP eq.
- The fixed points of the WP eq. are realized in the $\tau \rightarrow \infty$ limit along the GFERG flow

RG flow around fixed point

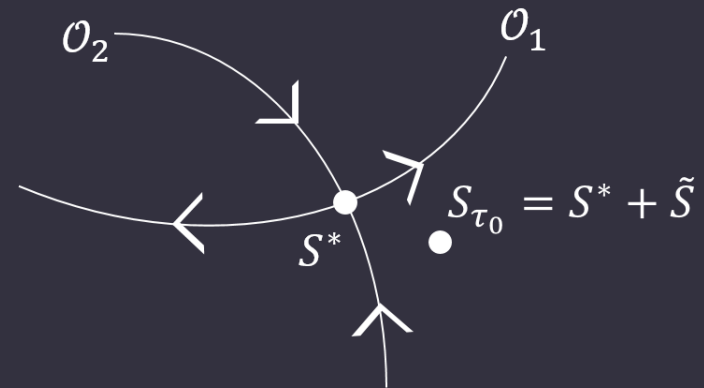
- Let us study **RG flow structure** around a fixed point after enough long time
- We set $\tau = \tau_0 + \delta\tau$ ($\tau_0 \gg 1$) and $S_{\tau=\tau_0} = S^* + \tilde{S}$
expansion parameters : $\lambda_0 := e^{-(D-2+\eta)\tau_0}, \tilde{S} \ll 1$
- Solution to the GFERG eq. up to the leading order

$$S_\tau = S^* + \sum_A \left(e^{x_A \delta\tau} \tilde{c}(0) - \lambda_0^{n_{\min}-1} e^{-(n_{\min}-1)(D-2+\eta)\delta\tau} c_A(f_a^{a_1, \dots, a_n}) + \text{const.} \right) \mathcal{O}_A$$

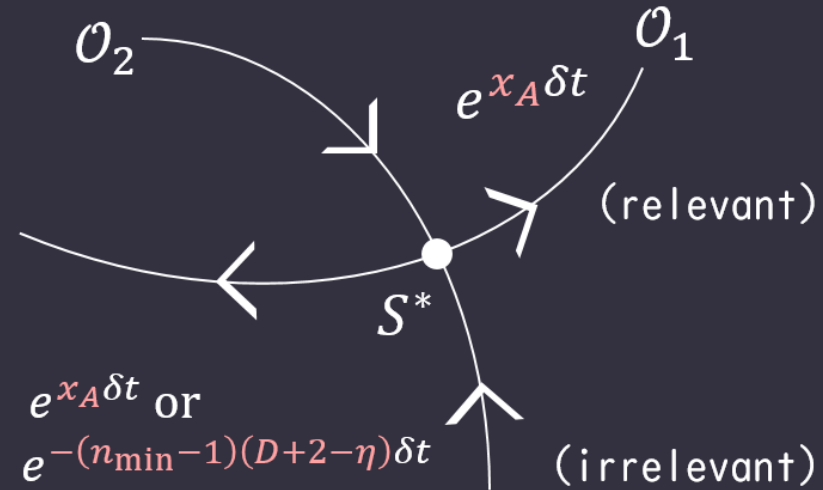
$\mathcal{O}_A$: eigen-operators of the linearized WP eq.
around S^*

x_A : eigenvalue of $\mathcal{O}_A$

$n_{\min}$: minimum order of the non-linear terms
in the GF eq



Scaling dimension



- Scaling Dimension d_A

$$d_A = \max(x_A, -(n_m - 1)(D - 2 + \eta))$$

$$x_A \geq 0 \text{ (relevant or marginal)} \implies d_A = x_A$$

$$x_A < 0 \text{ (irrelevant)} \implies d_A = -\min(|x_A|, (n_m - 1)(D - 2 + \eta))$$

- d_A of relevant or marginal operators are x_A
 $\implies$ critical exponents are the same as those of the WP eq.

Non-linear Sigma model

- Let us illustrate our results with
Non-linear sigma model in $4 - \epsilon$ dimensions
- Lagrangian

$$\mathcal{L} = \frac{1}{2g} (\partial_\mu \phi_i)^2, \quad (\phi_i)^2 = 1$$

- Its IR physics is described by
Wilson-Fisher fixed point in $4 - \epsilon$ dim.

$$S_{\text{WF}}^* = \int_x \frac{1}{2} (\partial_\mu \phi_i)^2 + \frac{m_*^2}{2} \phi_i^2 + \frac{\lambda_*}{8} (\phi_i^2)^2 + O(\epsilon^2)$$

$$m_*^2 := -\frac{\epsilon N + 2}{4N + 8}, \quad \lambda^* := \epsilon \frac{8\pi^2}{N + 8}$$

GFERG of NL Sigma Model

- Gradient flow equation

$$\partial_\tau \varphi_i = \partial_x^2 \varphi_i - (\varphi_j \partial_x^2 \varphi_j) \varphi_i$$

[H. Makino, H. Suzuki 1410.7538]

- GFERG equation

$$\partial_\tau e^{-S_\tau[\Phi]} = (\text{WP part})$$

$$\begin{aligned} & -\lambda^2(\tau) \int_x \frac{\delta}{\delta \phi_i(x)} \left(\left(\phi_j(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_j(y)} \right) \right. \\ & \left. \times \partial_x^2 \left(\phi_j(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_j(y)} \right) \left(\phi_i(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_i(y)} \right) \right) e^{-S_\tau[\Phi]} \end{aligned}$$

Fixed point, Scaling dimension

- Anomalous dimension at WF fixed point

$$\frac{\eta}{2} = \frac{N+2}{(N+8)^2} \frac{\epsilon^2}{2} + O(\epsilon^3)$$

$$\rightarrow \lambda(\tau) \sim \exp\left(-\tau\left(2 - \epsilon + \frac{N+2}{(N+8)^2} \epsilon^2\right)\right) \rightarrow 0 \quad (\tau \rightarrow \infty)$$

- Scaling dimensions

$$- \mathcal{O}_A = (\phi_i^2)/2 \rightarrow d_{\phi_i^2} = \max(2 - \epsilon, -2(D - 2 + \eta)) = 2 - \epsilon > 0$$

$$(\text{critical exponent } \nu^{-1} := d_{\phi_i^2} = 2 - \epsilon)$$

$$- \mathcal{O}_A = (\phi_i^2)/2 - (\phi_i^2)^2/8 \rightarrow d_A = \max(-\epsilon, -2(D - 2 + \eta)) = -\epsilon < 0$$

Conclusion

- Studied the fixed point structure of the GFERG eq. associated with the general gradient flow for scalar fields
- Showed that it is almost the same as that of the Wilson–Polchinski eq.
- Discussed that the GFERG eq. has a similar RG flow structure around a fixed point to the WP eq.

Thank you!

Back up

RG flow around fixed point

- Let us study RG flow structure around a fixed point after enough long time
- We set $\tau = \tau_0 + \delta\tau$ ($\tau_0 \gg 1$) and $S_{\tau=\tau_0} = S^* + \tilde{S}$
expansion parameters : $\lambda_0 := e^{-(D-2+\eta)\tau_0}, \tilde{S} \ll 1$
- Solution to the GFERG eq. up to the leading order

$$S_\tau = S^* + \sum_A \left(e^{x_A \delta\tau} \tilde{c}(0) - \lambda_0^{n_m} (e^{-n_m(D-2+\eta)\delta\tau} - 1) \frac{g^A}{x_A + n_m(D-2+\eta)} \right) \mathcal{O}_A$$

$\mathcal{O}_A$: complete set

$$\sum_A \tilde{c}(0) \mathcal{O}_A := \tilde{S} \quad n_m: \text{minimum power of nonlinear term in the GF eq}$$

$$\sum_A g^A \mathcal{O}_A := e^{S^*} \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_a^{i_1, \dots, i_{n_m}} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right)^{n_m} e^{-S^*[\Phi]} \right.$$

